

Empirical Proof Record: VALIDATED

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Proof ID: fae2098bf5faad4b1d9371434996a3ff...

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1. Claim

Across 8 real FRED series, the substrate recovers the true temporal direction (vs the exact reversal) at rate $\geq 65\%$ via a consistent path-dependent irreversibility signature. An order-blind reader is 0.5 by construction (identical value multiset).

- **Operationalisation:** For each series, take the last 48 values (z-scored); stream them FORWARD and REVERSED through the 'entity' target as bare-number stimuli (no direction words). For each path-dependent numeric observable in CycleResult.raw, compute a per-direction summary; the substrate 'votes' forward = the direction with the larger summary. `arrow_detection_rate` = sign-consistency of that vote vs the true arrow, max over observables (sign-agnostic: a flipped but consistent vote still counts as detection).
- **Threshold:** `arrow_detection_rate` ≥ 0.65 proportion
- **H0:** H0: rate ≈ 0.5 — the substrate is arrow-blind through the numeric channel (no consistent forward/reversed asymmetry)
- **H1:** H1: rate $\geq 65\%$ — the substrate feels the arrow of time via its path-dependent entropy production

2. Verdict

VALIDATED — observed 1.0

arrow-detection rate 1.000 on 8 real series (best observable: `thermo_bridge_state.layer1.dominant_signature.u`)
order-blind baseline = 0.500 by construction; Wilson 95% CI [0.676, 1.000]

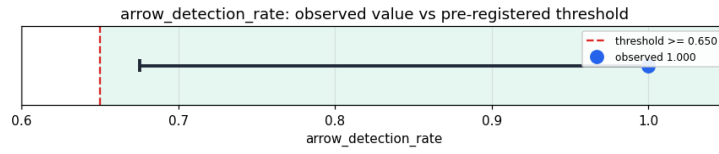


Figure 1: arrow_detection_rate – observed value with Wilson 95% CI.

3. Evidence

Pillar	Statistic	Value	95% CI	Library
path_dependent_irreversibility	arrow_detection_rate	1.0	(0.6756, 1.0000)	statsmodels 0.1

4. Pre-registration

- Registered at: 2026-05-23T23:16:46.399267+00:00
- Config hash: 890acd0a378abdf02e5e9878d430bb2b...
- Data hash: 0fce9b79cbca7980faeb297bc3f6d3ac...

5. Signature

58c23b94d3ccc3941066080c0c4b2ce077298b68566ba0c61e7a4bc7eed7beeb